

# JOSHUA HIMMENS

Engineering Physics Student

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## EDUCATION

University of British Columbia

Vancouver, BC, Canada

*Bachelor of Applied Science* in Engineering Physics

Sep 2023 – May 2028

- Capstone projects: Developed a fully robotic ML-controlled napkin folding machine, designed an audible phased array sonar system.
- Specialized in entrepreneurship, machine learning, quantum computing, robotics, and embedded systems.

## WORK EXPERIENCE

AltaML

Edmonton, AB, Canada (remote)

*Associate Machine Learning Developer* (full-time)

Jan 2025 – Apr 2025

- Developed machine learning models using scikit-learn to save the Government of Alberta >\$4 million annually.
- Developed novel evaluation metrics to minimize regional bias and maximize transferability of models.
- Built Azure ML pipelines for automated model training and deployment, serving predictions via REST APIs integrated with SQL databases and Blob Storage.
- Implemented Uber's H3 hexagonal spatial indexing to balance geographic accuracy with user privacy in an infrastructure planning tool.

TRIUMF

Vancouver, BC, Canada

*ATLAS Deep Learning Research Student*

May 2024 – Sep 2024 (full-time)

Sep 2024 – Feb 2025 (part-time, 10 hrs/week)

- Developed panoptic segmentation models for the ATLAS detector using the PointNet ML framework with wandb, TensorFlow, Keras.
- Used CERN's grid computing to parallelize compute across thousands of nodes.
- Worked independently to develop models using cutting-edge transfer learning approaches.
- Implemented models into production using NVIDIA Triton.

UBC Orbit Satellite Design Team

Vancouver, BC, Canada

*Command and Data Handling (CDH) Lead and Firmware Developer* (part-time, 10 hrs/week)

Oct 2023 – Present

- Led the CDH team to develop software to meet mission and testing objectives from ESA (European Space Agency) for the ALEASAT project.
- Managed a team of 10 firmware developers, with over \$15 000 worth of technical infrastructure.
- Developed mission testing, function testing, and acceptance testing procedures to meet ESA and ECSS standards.
- Programmed device drivers and electrical ground support equipment (EGSE).

*Embedded Firmware Developer* (part-time, 10 hrs/week)

Jan 2023 – Present

- Programmed device drivers, electrical ground support equipment (EGSE).
- Developed the ALEASAT Avionics Test Bench (FlatSat).
- Worked on the ALEASAT onboard computing system using FreeRTOS on a TMS570 microcontroller.

## PUBLICATIONS AND PRESENTATIONS

Machine Learning for Robotic Napkin Folding at 2026 IEEE Vancouver Section AGM

Mar 2026

- Placed 3rd. Explored various policies on SO-101 arms.

Developing Machine Learning Techniques for Particle Flow in the ATLAS Experiment at Canadian Astroparticle Physics Summer Student Talks Competition (CASST 2024)

Jun 2024

- Placed 2nd of 44 presentations.

JetPointNet: A Machine Learning Approach to Cell-to-Track Attribution in the ATLAS Experiment at Canadian Undergraduate Physics Conference

Oct 2024

3D Particle Flow in the ATLAS Calorimeter: How to Train Your Model at TRIUMF Science Week

May 2024

## HONORS AND AWARDS

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### Hector J. MacLeod Award

Issued by *IEEE Vancouver Section*

Apr 2024

- Awarded to students who have demonstrated pioneering efforts in their chosen field of study and attained high scholastic honours.

### Erich Vogt First Year Summer Research Experience (FYSRE) Award

Issued by *UBC Department of Physics and Astronomy*

May 2024

- Awarded to promising students in physics for a 4-month research placement.

### Alberta Centennial Award and Alberta Premier's Citizenship Award

Issued by *Government of Alberta*

May 2023

- Awarded for outstanding community service.

### Calgary Flames Foundation Community Involvement Scholarship

Issued by *Education Matters*

Aug 2023

- Awarded for community involvement.

### Julia Turnbull Leadership Award for exceptional community service

Issued by *The Calgary Foundation*

Jul 2023

- Awarded for exceptional community service.

## LEADERSHIP AND ADVOCACY

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### Chair

UBC IEEE Aerospace Electrical Systems Student Branch Chapter

Vancouver

Aug 2025 – Present

- Served as **Technical Program Chair** for the Student Aerospace Summit, September 2025.
- Organized events to unify aerospace-interested students across UBC.
- Worked with members across Canada to develop opportunity lists for students and recent graduates.
- Shared McDonald Institute opportunities with eligible HQP in BC and Alberta.

### Advisory Team Member

Child Rights Connect

Geneva, Switzerland (remote)

Jan 2021 – Dec 2023

- Provided guidance to UN delegations on communication strategies for high-level rights goals.
- Presented to governments and consulted on international initiatives to support the UN Convention on the Rights of the Child.
- Created and edited content on human and child rights in the Americas.
- Attended the 3rd Pan American Child and Youth Forum in Cartagena, Colombia with the Government of Canada.

## TECHNICAL SKILLS

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- **Machine Learning:** TensorFlow, Keras, PointNet, Weights and Biases (wandb), ONNX, LeRobot, Runpod, Triton
- **Embedded Programming:** Experienced with FreeRTOS on TMS570, RP2040, STM32, Xilinx and Altera FPGAs, KiCAD
- **Software Development:** Python, Rust, C, C++, Java, MATLAB, Bash, Git
- **Quantum Computing:** Qiskit, QASM, Pasqal, QCNNs, VQCs
- **Particle Physics:** ROOT, CERN grid computing, Athena

## ADDITIONAL EXPERIENCE

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- **Quantum School for Young Students** (University of Waterloo and Institute for Quantum Computing): Participant in intensive quantum computing program
- **Introduction to Quantum Computing** (IBM): Participant in 8-month course using IBM's quantum infrastructure
- **Scientific Computing with Python certification** (freeCodeCamp): 300 hours of training
- **GENQ Global Hackathon (Geneva, 2025 and Calgary, 2024)** (Quantum AI Ventures): Build QCNNs and QUBO solvers for quantum machine learning tasks